

EXP 29: Prolog Program for Forward Chaining

Name: Y. Roshitha (192424400)

Aim:

To develop a Prolog program that demonstrates forward chaining, so that new facts are derived from existing facts and rules until no more facts can be added.

Algorithm:

- Step 1: Start the program.
- Step 2: Store the initial facts in the knowledge base.
- Step 3: Examine all available rules.
- Step 4: Check whether a rule can derive a new fact.
- Step 5: Add the newly derived fact to the knowledge base.
- Step 6: Continue applying applicable rules.
- Step 7: Repeat until no new facts can be generated.
- Step 8: Display the derived facts.
- Step 9: Stop the program.

Source Code:

```
% Program for Forward Chaining
```

```
:- dynamic(fact/1).
```

```
fact(has_feathers).
```

```
fact(lays_eggs).
```

```
fact(can_fly).
```

```
rule(bird) :-
```

```
    fact(has_feathers),
```

```
    fact(lays_eggs).
```

```
rule(flying_bird) :-
```

```
    fact(bird),
```

```
    fact(can_fly).
```

```
forward_chain :-
```

```
    rule(X),
```

```
    \+ fact(X),
```

```
    !,
```

```
    assertz(fact(X)),
```

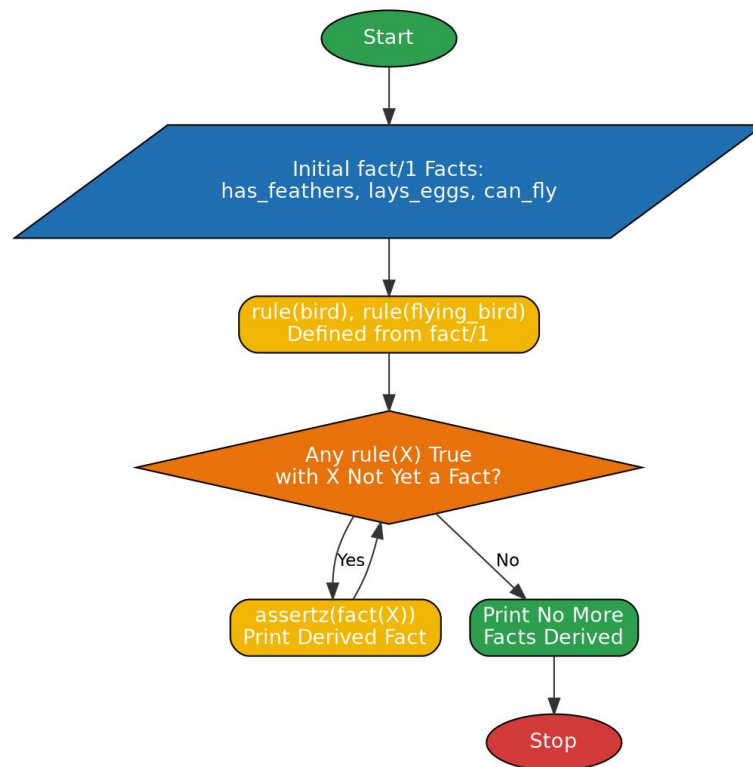
```
    format("Derived new fact: ~w~n", [X]),
```

```
    forward_chain.
```

```
forward_chain :-
```

```
    format("No more facts can be derived.~n").
```

Flowchart:



Output:

```
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?- cd('C:/Users/yanar/Documents/CSA1717').
true.

?- [forward].
true.

?- forward_chain.
Derived new fact: bird
Derived new fact: flying_bird
No more facts can be derived.
true.
```

Result:

Thus the Prolog program for forward chaining was written, executed successfully and the output was verified.